```
clc;
clear;
close all;

% DESIGN AND ANALYSIS OF FIR FILTERS
% Parks-McClellan vs Frequency-Sampling

Fs = 10000;             % Sampling frequency (Hz)
N = 40;                 % Filter order
Fpass = 1800;           % Passband edge (Hz)
Fstop = 2300;           % Stopband edge (Hz)

% Normalized frequencies (0 to 1, where 1 = Nyquist)
F = [0 Fpass Fstop Fs/2] / (Fs/2);
A = [1 1 0 0];

% 1. Parks-McClellan Equiripple Method
b_pm = firpm(N, F, A);

% 2. Frequency-Sampling Method
b_fs = fir2(N, F, A);

% Frequency response
[H_pm, freq] = freqz(b_pm, 1, 8192, Fs);
H_fs = freqz(b_fs, 1, 8192, Fs);

% Magnitude response in dB
Mag_pm = 20*log10(max(abs(H_pm), 1e-4));
Mag_fs = 20*log10(max(abs(H_fs), 1e-4));

% Phase response in degrees (wrapped)
Phase_pm = angle(H_pm)*180/pi;
Phase_fs = angle(H_fs)*180/pi;

% Create figure similar to reference image
figure('Color','w','Position',[100 50 900 700]);

% GRAPH 1: Magnitude Response Comparison
subplot(2,1,1);

plot(freq, Mag_pm, 'b-', 'LineWidth', 1.5);
hold on;
plot(freq, Mag_fs, 'r--', 'LineWidth', 1.5);

grid on;
xlim([0 5000]);
ylim([-80 10]);

xticks(0:500:5000);
yticks(-80:20:0);

xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');

title('FIR Filter Magnitude Response Comparison (Order N = 40)');

legend('Parks-McClellan (Equiripple)', ...
    'Frequency-Sampling', ...
    'Location','northeast');

% GRAPH 2: Phase Response Comparison
subplot(2,1,2);

plot(freq, Phase_pm, 'b-', 'LineWidth', 1.5);
hold on;
plot(freq, Phase_fs, 'r--', 'LineWidth', 1.5);

grid on;
xlim([0 5000]);
ylim([-200 200]);

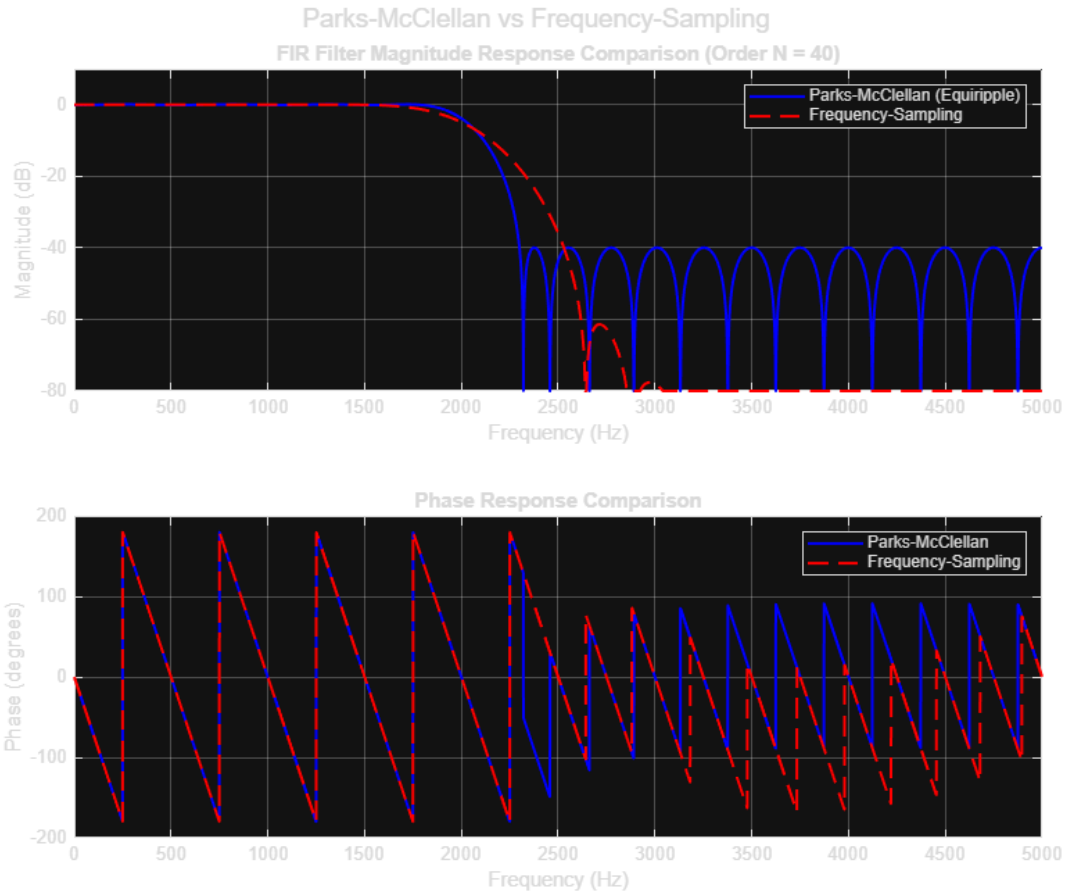
xticks(0:500:5000);
yticks(-200:100:200);

xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

title('Phase Response Comparison');

legend('Parks-McClellan', ...
    'Frequency-Sampling', ...
    'Location','northeast');

% Overall heading
sgtitle('Parks-McClellan vs Frequency-Sampling');
```



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